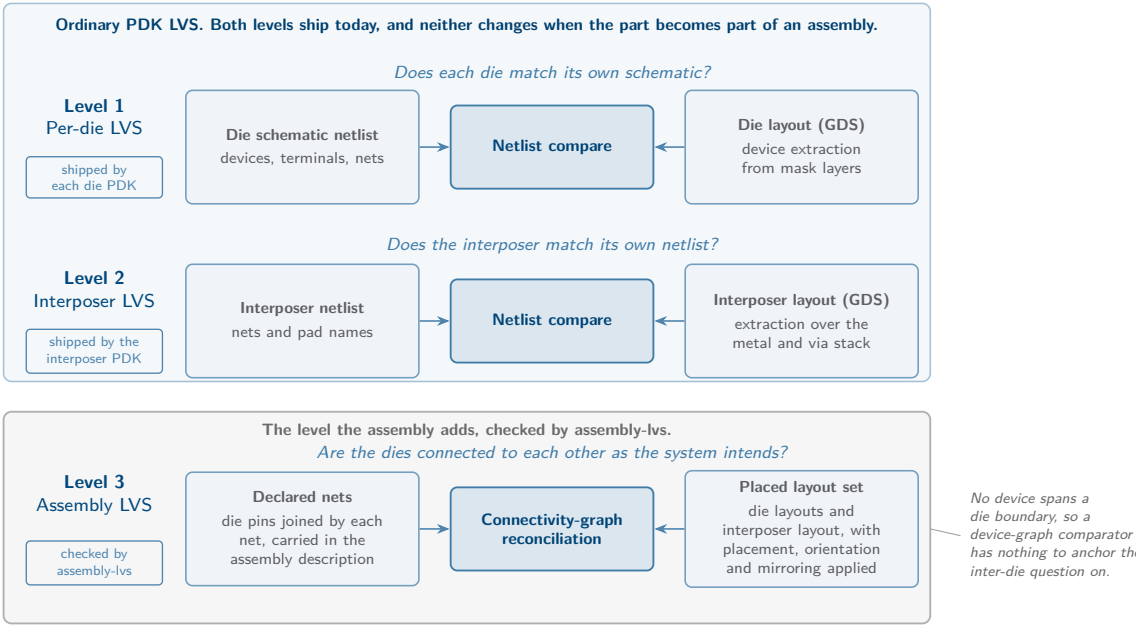

Ordinary PDK LVS. Both levels ship today, and neither changes when the part becomes part of an assembly.
Does each die match its own schematic?
Level 1
Per-die LVS
shipped by each die PDK
Die schematic netlist
devices, terminals, nets
Netlist compare
Die layout (GDS)
device extraction from mask layers
Does the interposer match its own netlist?
Level 2
Interposer LVS
shipped by the interposer PDK
Interposer netlist
nets and pad names
Netlist compare
Interposer layout (GDS)
extraction over the metal and via stack
The level the assembly adds, checked by assembly-lvs.
Are the dies connected to each other as the system intends?
Level 3
Assembly LVS
checked by assembly-lvs
Declared nets
die pins joined by each net, carried in the assembly description
Connectivity-graph reconciliation
Placed layout set
die layouts and interposer layout, with placement, orientation and mirroring applied
No device spans a die boundary, so a device-graph comparator has nothing to anchor the inter-die question on.